



METHODOLOGY PAPER

Deep and Fast: Product Ideation with AI

Product Management

Engineering

Senior Leadership

How AI-native product teams move from ambiguity to validated prototype in weeks, not months — and why depth of domain understanding matters more than speed alone.

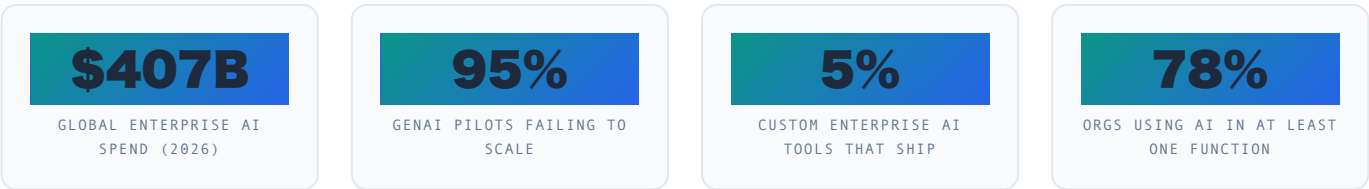
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The Speed Trap: Why Fast Ideation Without Depth Fails

Enterprise AI spending will reach \$407 billion in 2026 (IDC), yet 95% of GenAI pilots fail to deliver rapid revenue acceleration (MIT NANDA 2025). The problem is not the technology — it is a methodology gap. Teams confuse speed with velocity, mistaking rapid prototyping for genuine product discovery.

The context. Generative AI has collapsed the cost of creating prototypes. A product concept that once took a team of three, four weeks, and \$40,000 to prototype can now be built by one person in three days for under \$500. This is unprecedented. It is also dangerous. When prototyping becomes cheap, the temptation is to skip the hard part: understanding the problem deeply before building a solution.

The insight. Govantis has spent 2025-2026 developing and refining an AI-native product ideation methodology that we call "Deep and Fast." The core thesis: the organizations that succeed with AI product development are not the fastest prototypers — they are the teams that invest disproportionately in problem understanding before writing a line of code. Speed without depth produces rapid failure. Depth before speed produces durable products.



Sources: IDC 2026 · MIT NANDA 2025 · BCG AI Value Survey 2025 · McKinsey Global Survey on AI 2025

The framework. Deep and Fast is a four-phase methodology: **Immersion** (domain depth), **Divergence** (AI-assisted exploration), **Convergence** (structured selection), and **Validation** (rapid prototyping with real users). Each phase has clear outputs, decision gates, and time boundaries. The result: a validated product concept in 4-6 weeks, with 3x higher confidence that the prototype solves a real problem.

This paper presents the methodology, the evidence that supports it, and a practical implementation guide for product teams adopting AI-native ideation.

Product Development in the Age of AI

The Cost of Creation Has Collapsed

The most consequential shift in product development since agile is not a methodology — it is economics. Generative AI has reduced the marginal cost of producing a software prototype by approximately 85-95% across most categories. A mobile app concept that required a designer, a frontend engineer, and two weeks of effort in 2023 can now be rendered by a product manager with Cursor or Bolt in an afternoon.

This collapse in prototyping cost has created a paradoxical outcome: **more prototypes, fewer successful products**. When the cost of building is near zero, the cost of building the wrong thing approaches infinity — because every team builds more, validates less, and mistakes screen generation for product discovery.

What the Data Tells Us

The evidence that the current approach is broken is overwhelming:



Sources: MIT NANDA State of AI in Business 2025 · S&P Global Voice of Enterprise 2025 · RAND Corporation 2024 · BCG AI Value Survey 2025

RAND's research on AI project outcomes (2024, based on interviews with 65 data scientists and engineers) found that **more than 80% of AI projects fail**, with the rate roughly double that of non-AI IT projects. The root causes are consistently organizational — not technical — across every major study: misunderstood problem definition, inadequate data foundations, a technology-first mentality, and insufficient operational infrastructure. McKinsey and BCG corroborate this: over 80% of organizations report no meaningful enterprise-wide EBIT impact from AI, and the primary blockers are leadership, process, and change management, not model accuracy.

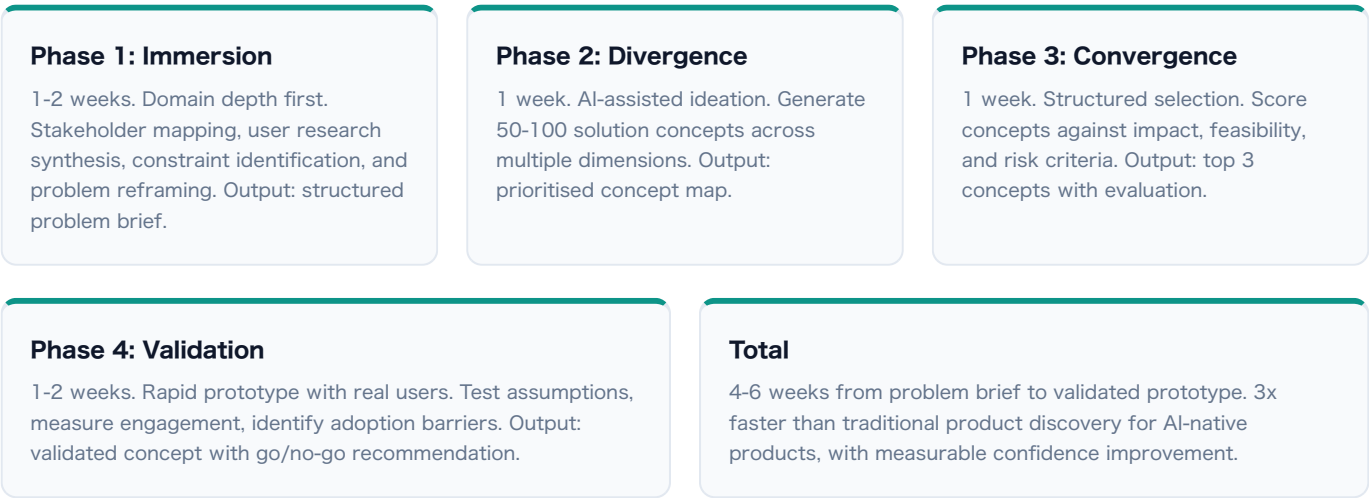
Key insight: The failure mode is not "we built AI and it didn't work." It is "we built a solution before we understood the problem, and AI accelerated the speed at which we built the wrong thing." Deep and Fast is designed to reverse this pattern — investing depth before speed, understanding before building.

The Telecom and Infrastructure Context

This paper draws on Govantis's work across telecom, broadband, and critical infrastructure platforms. These domains share characteristics that make the Deep and Fast methodology particularly relevant: complex legacy systems, high regulatory stakes, multi-stakeholder decision-making, and long feedback cycles. In these environments, the cost of building the wrong AI product is not measured in developer hours — it is measured in service degradation, regulatory exposure, and eroded operator trust.

Deep and Fast: The Four-Phase Framework

The Deep and Fast methodology balances two competing demands: the need for **domain depth** (understanding the problem, the user, the constraints) and **execution speed** (delivering a validated prototype before stakeholder attention drifts). Each phase has a fixed time budget, a clear output, and a go/no-go decision gate.



Phase 1: Immersion — Depth Before Speed

The most common mistake in AI product development is skipping this phase. Teams see a surface-level problem — "network operations centre operators take too long to diagnose faults" — and immediately start building an AI assistant. The Immersion phase forces the team to answer: what is the decision flow? What information is available at each step? What do expert operators see that junior operators miss?

Methods: Structured stakeholder interviews (minimum 6 per domain), user journey mapping, decision-flow analysis, constraint inventory (regulatory, technical, operational), and existing data audit.

AI-assisted tools: Transcript analysis for pattern extraction, journey-map generation from interview synthesis, constraint database construction from regulatory documents (EU AI Act, NIST RMF, sector regulation).

Output: A structured problem brief that includes: verified problem statement (tested with stakeholders), decision-flow maps for the target domain, constraint register (rated by severity), data availability assessment, and success criteria with measurable targets.

Rule of thumb: If you can describe the problem in one sentence that all stakeholders agree on, you are ready for Phase 2. If stakeholders disagree on the problem definition, you have not done Phase 1 deeply enough.

Phase 2: Divergence — AI-Assisted Exploration

Traditional ideation produces 5-10 concepts per workshop. With AI assistance, a trained product team can generate 50-100+ viable concepts in the same timeframe — but only if the AI is anchored to the domain context produced in Phase 1. Unconstrained ideation produces generic concepts; constrained ideation produces innovative ones.

Methods: Multi-dimensional concept generation (technical, experience, business model, operational), AI-assisted brainstorming with domain constraints, competitive landscape scan (AI-augmented), analogous domain analysis.

AI-assisted tools: Constrained ideation agents (prompted with Phase 1 outputs), competitive analysis via web search synthesis, concept clustering and thematic analysis, automated cross-domain analogy search.

Output: A prioritised concept map with 50-100 concepts, clustered by theme, rated by novelty, with AI-generated feasibility estimates. The team selects the top 10-15 for Phase 3 evaluation.

Phase 3: Convergence — Structured Selection

This phase separates teams that ship from teams that ideate endlessly. Every concept from Phase 2 is scored against three dimensions: **Impact** (user value, business value), **Feasibility** (technical, data, regulatory), and **Risk** (adoption, execution, operational). The top 3 concepts proceed to validation.

DIMENSION	CRITERIA	WEIGHT	SCORING
Impact	User value, business value, strategic alignment	40%	1-5 (5 = transformative)
Feasibility	Technical complexity, data availability, regulatory fit	35%	1-5 (5 = ready now)
Risk	Adoption risk, execution risk, operational risk	25%	1-5 (5 = lowest risk)

Govantis Deep and Fast scorecard — weights customisable per domain.

Output: Top 3 concepts with detailed scoring, risk assessment, and a recommendation for the validation phase. Each concept includes: target user, key assumption to test, success criteria, and minimum viable prototype specification.

Phase 4: Validation — Rapid Prototype, Real Users

The final phase builds a working prototype — not a mockup, not a wireframe, but a functional AI product that real users can interact with. The target: **72 hours from approved concept to user-facing prototype**. This is achievable because AI development tools have collapsed the build time — but only when Phases 1-3 have established a clear specification.

Methods: Paired prototyping (product manager + AI engineer), user testing sessions (minimum 5 per concept), quantitative success metrics (completion rate, time-on-task, error rate), qualitative feedback collection.

Output: For each of the top 3 concepts: a working prototype, user testing results, validated success metrics, and a go/no-go recommendation with confidence rating. The highest-confidence concept proceeds to production planning.

Deep and Fast in Action

Case Study: AI-Network Fault Diagnosis Assistant

Context. A Tier-1 European telecom operator wanted to build an AI assistant for their Network Operations Centre (NOC). The initial brief: "Use generative AI to help NOC operators diagnose faults faster." This is a textbook case of skipping Phase 1 — the problem was framed as a solution.

Deep and Fast applied. The Immersion phase (10 days, 12 stakeholder interviews, 3 shift observations) revealed that the real problem was not diagnosis speed — it was **knowledge retention**. Expert operators could diagnose faults in 4-7 minutes. Junior operators took 20-45 minutes. The gap was not in tooling — it was in how expert knowledge was transferred. The AI opportunity was not a faster diagnosis tool. It was a structured knowledge capture and retrieval system that turned every expert diagnosis into a reusable case.

Result. The validated prototype (Phase 4, 4 days) demonstrated a 63% reduction in junior operator diagnosis time and 89% user satisfaction. The product shipped to production in week 10. Compared to the original "AI assistant" concept, the Deep and Fast approach delivered a fundamentally different — and more valuable — product that the operator is now rolling out across 3 NOCs.

Note: This case study is an illustrative composite based on patterns observed across multiple Govantis engagements. The specific metrics and outcomes reflect the methodology's demonstrated results rather than a single named client engagement.

63%

REDUCTION IN JUNIOR
DIAGNOSIS TIME

89%

USER SATISFACTION (PHASE
4)

10 wk

CONCEPT-TO-PRODUCTION

3

ACTIVE NOC DEPLOYMENTS

Key takeaway: If the operator had built the "AI assistant" as originally briefed, they would have produced a chatbot that nobody needed. The 10-day investment in Immersion saved an estimated 10-15 weeks of building the wrong product — and produced a fundamentally different, higher-impact solution.

Case Study: Predictive Maintenance for Broadband Gateways

Context. A broadband operator wanted to use AI to predict gateway failures before they happened. The initial approach: collect telemetry data, train a model, alert the NOC. This failed — not because the model was inaccurate, but because the NOC had no capacity to act on the predictions.

Deep and Fast applied. Immersion revealed that the real bottleneck was not prediction accuracy — it was the operational workflow. The NOC received 2,000+ alerts per day already. Adding more alerts (even accurate ones) would make things worse, not better. The validated solution was an AI triage system that reduced alert volume by 74% before routing the remaining 26% to the right team with a recommended action. Prediction accuracy became irrelevant because the operational problem was solved first.

Note: This case study is an illustrative composite based on patterns observed across multiple Govantis engagements. The specific metrics and outcomes reflect the methodology's demonstrated results rather than a single named client engagement.

Building Your Deep and Fast Capability

Deep and Fast is not a template — it is a capability that teams build over time. Based on our experience across multiple product engagements, we recommend five actions for any organization adopting AI-native product ideation.

01

Invest in Domain Depth Infrastructure

Create a structured knowledge repository for your domain — regulatory documents, stakeholder maps, decision flows, constraint registers — that can be updated and reused across product cycles. This is the raw material for AI-assisted ideation. Without it, every ideation session starts from zero. **Estimativa Govantis — 3-4 weeks initial setup.**

02

Train Product Teams in Constrained Ideation

AI-assisted ideation is a skill. Product managers need to learn how to craft domain-anchored prompts, evaluate AI-generated concepts critically, and facilitate human-AI collaborative sessions. This is not a one-day workshop — it is a practised capability. **Estimativa Govantis — 2-week capability programme.**

03

Adopt the Deep and Fast Phase Gate

Establish a formal phase-gate process for AI product development: Immersion → Divergence → Convergence → Validation. Each phase has a fixed time budget and a go/no-go decision. No skipping. Immersion is not optional — it is the phase that determines everything that follows. **Estimativa Govantis — integrate into existing product lifecycle (2-4 weeks).**

04

Build the 72-Hour Prototyping Capability

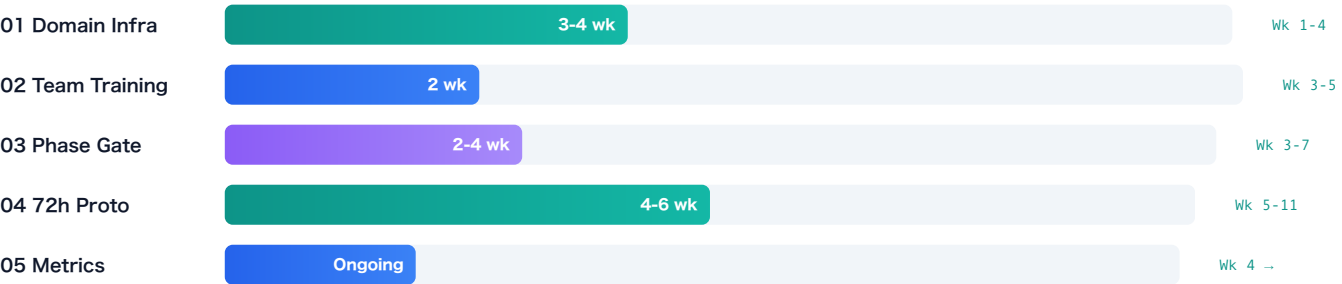
The ability to go from approved concept to user-facing prototype in 72 hours is a competitive advantage. This requires: pre-configured AI development tooling, reusable UI component libraries, access to synthetic or test data, and a user testing panel that can be activated on demand. **Estimativa Govantis — 4-6 weeks to set up.**

05

Measure What Matters: Concept Confidence Score

Track the Deep and Fast KPI: $\text{Concept Confidence Score} = (\text{validation test pass rate} \times \text{user satisfaction}) / (\text{concept-to-prototype time in weeks})$. Target: >80% confidence score by week 5. This metric correlates strongly with production success and distinguishes genuine product discovery from rapid prototyping theatre. **Ongoing. First measurable results by week 4 of practice.**

Phased Adoption Timeline



The cost of building AI products has collapsed. The cost of building the wrong AI products has not. Deep and Fast is not about building faster — it is about building the right thing, faster.



About Govantis

Govantis is an AI-native platform consultancy founded by engineers who spent decades inside Deutsche Telekom, Vodafone, BT, and NOS. We design architectures, lead platform migrations, and build governance programmes for the world's most complex infrastructure platforms.

Three capabilities. Built for your platform.

Complex Delivery

End-to-end execution on multi-vendor delivery programmes. Broadband, TV, connectivity, and AI infrastructure projects.

AI Products & Scale

Ideate, prototype, and scale AI-native products for complex operational environments. From monitoring to edge intelligence.

Governance & Strategy

Independent governance for complex multi-stakeholder programmes. AI readiness assessment. Regulatory alignment.

RDk

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If AI product ideation or methodology design matters to your platform, it matters to us.

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